

NITESH SOLANKI

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GitHub | Leetcode | CodeChef | **Roll No:** 23U01049

EDUCATION

Indian Institute of Information Technology, Bhopal

Bhopal, India

BTech in Electronics & Communication Engineering; CGPA: 7.41/10

2023 – 2027

- **Relevant Coursework:** Machine Learning (FML/Ops), Deep Learning, Data Structures & Algorithms.
- **Achievement:** 95+ percentile (among 12+ lakh students) in JEE Mains 2023.

SKILLS

Problem Solving: Solved 200+ Data Structures & Algorithms questions across platforms (LeetCode, GFG, Coding Ninjas).

Languages: C/C++, Python, HTML, CSS, JavaScript, SQL.

Machine & Deep Learning: Machine Learning, Deep Learning (CNNs, RNNs, LSTMs, Transformers, GANs), Computer Vision, NLP.

Data Science & Libs: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, PyTorch, Hugging Face.

Tools & Frameworks: Flask, Streamlit, React, MLflow, Docker, FastAPI, Git, GitHub, MySQL, Jupyter, Colab.

PROJECTS

Portfolio Optimizer MLOps | *FastAPI, Airflow, MLflow, Docker, DVC, LSTM*

[GitHub](#)

- Engineered LSTM-based stock forecasting models with Fast Fourier Transform (FFT) filtering for short-term accuracy.
- Containerized workflows with FastAPI and orchestrated Airflow pipelines for retraining and drift monitoring.
- Delivered up to 60% higher simulated returns using reproducible ML workflows via MLflow, DVC, and Docker.

Agentic Voice Assistant | *STT, LLM, RAG, TTS, ChromaDB, FastAPI, Streamlit*

[GitHub](#)

- Built an end-to-end voice assistant using Speech-to-Text, LLM-RAG, and Text-to-Speech for automated customer support.
- Implemented semantic retrieval with ChromaDB and vector embeddings for accurate, context-aware responses.
- Developed modular FastAPI backend services coupled with a Streamlit interface for robust and scalable audio pipelines.

Consumer Segmentation System | *K-Means, FastAPI, Streamlit, Pandas*

[GitHub](#)

- Built a customer segmentation pipeline using K-Means clustering and robust data preprocessing (encoding, scaling).
- Developed scalable FastAPI REST APIs for real-time segment prediction and automated model retraining workflows.
- Created a Streamlit dashboard offering actionable customer insights and user-driven predictions.

Student Health Risk Predictor | *Machine Learning, Streamlit, Python*

[GitHub](#)

- Built a machine learning model to accurately predict student health risk using structured dataset features.
- Implemented comprehensive data preprocessing, feature engineering, and an efficient model training pipeline.
- Developed an interactive Streamlit application for real-time risk prediction and data visualization dashboards.

LEADERSHIP & EXTRACURRICULARS

Technical Leadership

Project Lead

Mentorship & Deployment

- Led the conceptualization, development, and deployment of the AI-driven Portfolio Optimization System.
- Mentored peers on best practices in Machine Learning, Data Science, and modern full-stack web development.

Sports Achievements

University Teams (Table Tennis & Cricket)

- **Table Tennis:** Selected in NSO table tennis (5/100 acceptance ratio); active participant in university-level events.
- **Cricket:** Runner up in Inter-IIIT Competition; key contributor across multiple inter-university matches.